

On the mismatch between measured and perceived sleep quality

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ABSTRACT

Sleep quality has a significant impact on human's physical and mental well-being. While there exist several definitions of what makes a good quantity and quality of sleep, such measures are not always in agreement with each other. Further, they often do not match with the human perception of a restorative sleep. In this paper, we investigate the correlation between *objective* sleep quality measurements – which can be sensed using wearable devices – and the sleeper's *subjective* – i.e., self-reported – sleep quality. We analyze the M2Sleep dataset, which contains sleep data of 16 participants over one month, and show that only few sensor-based measures (sleep duration, skin temperature, tonic component of electrodermal activity, acceleration skewness and kurtosis) and personality-based features (extroversion, agreeableness, PSQI score) correlate with self-reported sleep quality. These results may inform the design of future studies on sleep quality assessment.

CCS CONCEPTS

• **Human-centered computing** → **Ubiquitous and mobile computing systems and tools.**

KEYWORDS

Sleep Quality; Subjective; Objective; Wearable Sensors; Correlation Analysis

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1 INTRODUCTION

Sleep behaviour is known to be instrumental for human well-being. On the short term, bad sleep quality negatively impacts concentration and causes headache and dizziness [16]. On the long term, it

favours the onset of serious conditions such as diabetes [5], cancer [23], weakened immune system [17], Parkinson's disease [13], and more.

Given the pivotal role of sleep for human physical and mental well-being, both research and industry have devolved significant attention to systems that are able to support and improve sleep quality. Several research-grade and commercial sleep tracking devices exist. Polysomnography (PSG) is considered the gold standard for sleep quality measurements. It uses Electroencephalogram (EEG), Electrocardiogram (ECG), Electromyography (EMG) and Electrooculography (EOG) to extract several sleep features such as, e.g., total sleep time and sleep efficiency. However, even when an accurate measurement method such as PSG is used, researchers observed a mismatch between measured and self-reported sleep quality [12]. This is arguably due to the fact that sleep quality is a complex construct that encompasses both “objective” and “subjective” aspects and summarizes “one's satisfaction of the sleep experience, integrating aspects of sleep initiation, sleep maintenance, sleep quantity, and refreshment upon awakening” [14].

With the recent widespread of wearable devices for monitoring sleep quality – which typically provide lower quality data than PSG – the issue of how accurate sleep quality measurements are and how well they match self-perceived sleep quality is exacerbated. The aim of this paper is to investigate this issue by comparing human self-reported scores of sleep quality with “objective” measurements provided by wearable devices. To this end, we leverage the M2Sleep dataset presented by Gashi et al. [8], which contains self-reported data and physiological signals collected from 16 participants over one month. We apply several statistical tests to this data and explore the correlation between self-reported sleep quality and features computed from physiological signals and other data.

Overall, our results show that often only a weak correlation between objective features (i.e., measured by sensing devices) and subjective (i.e., self-reported) measures of sleep quality can be found. Thereby, we outline that specific metrics measured both *before* and *during* sleep influence sleep quality, whereas we did not find any correlation between sleep quality and physiological properties measured after waking up. Lastly, we expose a correlation between certain personality traits with self-reported sleep quality, which may hint at the need to correct for self-report bias in sleep quality assessment studies.

2 RELATED WORK

Our work looks at the correlation between self-reported sleep quality and measures computed through (wearable) computing devices

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and is thus related to approaches that look at both subjective and objective measures of sleep quality. To quantify sleep behavior in a subjective way several questionnaires can be used. A popular one, used in many of the studies cited below, is the Pittsburgh Sleep Quality Index (PSQI) [6]. The PSQI evaluates sleep behavior on the past 30 days, and a higher PSQI total score indicates worse sleep quality. Some studies use also a reduced version of the questionnaire, cutting the original 19 questions to a few or just one.

Several studies investigated the correlation between subjective and objective measures of sleep quality [1, 3, 15, 18, 19]. Sano and Picard [19] explored the correlation between Electrodermal activity (EDA), sleep stages and sleep quality using data from six participants during one night. They discovered that the higher the number of EDA storms during the first quarter of the night, the higher the subjective sleep quality. Sajjadih et al. [18] showed that there is an association between higher heart-rate and poor sleep quality; Borazio and Van Laerhoven [3] that copious movements during the night correlate with poor self-reported sleep quality; and Kräuchi et al. [15] that the vasodilation of distal skin favours a rapid onset of sleep. Åkerstedt et al. [1] explored the correlation between subjective sleep quality, sleep efficiency and sleep stages in eight participants over eight night. They discovered that subjective sleep quality was strongly related to sleep efficiency, but no correlation with sleep stages was found. There are also external factors that can influence sleep quality. Casiraghi et al. [7] showed that the moon cycle can affect sleep. Furthermore, humans tend to have different sleep schedules on weekdays and weekends, as Zhang et al. [22] pointed out.

In this work, we build upon the approaches mentioned above and explore how physiological features – extracted *before*, *during* and *after* sleep – correlate with self-reported sleep quality. We further investigate how personality traits and other data influence the assessment of sleep quality.

3 DATASET, DATA PRE-PROCESSING AND FEATURE EXTRACTION

The analysis described in this paper relies on the M2Sleep dataset by Gashi et al. [8], which contains self-reports about sleep quality and physiological signals collected using the Empatica E4 wristband¹ from 16 participants over one month. The dataset includes demographic data, PSQI, Big Five Inventory (BFI) and Munich Chrono-Type Questionnaire (MCTQ), which the participants reported at the start of the data collection. It also contains daily self-reports about the time at which participants went to bed and woke up as well as the sleep quality level (5-point Likert scale from 1 to 5). If a participant took a nap during the day or woke up during the night, self-reports are available for each sleep segment. For all participants the dataset also includes sensor traces of Photoplethysmography (PPG), 3 axis Acceleration (ACC), Optical thermometer, Electrodermal Activity (EDA) data, collected using the on-board sensors of the E4 Empatica wristband. For two participants the dataset also contains sleep start, sleep end and sleep quality measurements computed by a MiBand² wristband, which is considered an objective tool to assess sleep [11].

¹<https://www.empatica.com/en-gb/research/e4/>

²<https://www.mi.com/global/mi-smart-band-5/>

To run our analysis we divide the sensor traces in the M2Sleep dataset in *before sleep*, *during sleep* and *after sleep* segments. We define *before sleep* as the time before the first self-reported sleep onset during a night; time *after sleep* as the time after the last wake up entry; and time *during sleep* as everything in-between³. We associate a sleep quality label to each *during sleep* segment and obtain: 10 score 1 labels (2%), 81 score 2 (16%), 220 score 3 (44%), 167 score 4 (34%); and 22 score 5 (4%). A score of 1, 2, 3, 4 and 5 corresponds to *very poor*, *poor*, *normal*, *good* and *very good* sleep quality.

To investigate the correlation between “objective” metrics and self-reported sleep quality labels, we extract different type of features: self-reports (from questionnaires), quantitative sleep metrics (sleep duration and number of awakenings), context metrics (day of the week, weekend and lunar phase), sensor-based features. For simplicity, we group these features in two categories: non-sensor-based and sensor-based features. The non-sensor-based features are: sleep-duration, number of awakenings, weekday, weekend, lunar phase, agreeableness (BFI_AGR), openness (BFI_OPEN), conscientiousness (BFI_CON), extraversion (BFI_EXT), neuroticism (BFI_NEU), participant age, average sleep duration (MCTQ), weekly sleep loss (MCTQ). We further select *sensor-based* features that, according to the literature [3, 15, 18, 19], are related to sleep quality. Table 1 provides the list of such features along with their description and the methods used to compute them. Since the amplitude of physiological signals may vary significantly across individuals, we normalized all the sensor-based features. We follow Zontone et al.’s normalization procedure [24], which basically subtracts the mean from each feature. Lastly, for each of the sensor-based features in each of the *before sleep*, *during sleep* and *after sleep* segment we further compute the following statistical features: sum, mean, median, mean absolute deviation (mad), minimum, maximum, mode, standard deviation (std), difference between maximum and minimum, ratio between maximum and minimum, skew, kurtosis, and 10th, 20th, 30th, 40th, 60th, 70th and 80th percentile.

4 RESULTS AND DISCUSSION

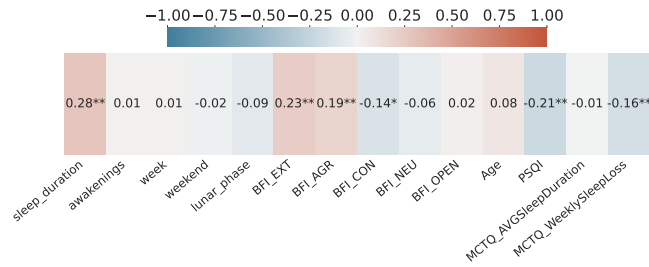
Using the data available in the M2Sleep dataset we first look at the agreement between the sleep quality score provided by commercial devices and the self-reported sleep quality. Specifically, two users wore during the whole study, in addition to the E4 wristband on the non-dominant hand, also a MiBand⁵ sleep monitoring device on the other hand. The MiBand gives a sleep quality score from 0 to 100, which we scaled to correspond to the 5 levels of the Likert scale used for the users’ self-reports. We then analyze how well the MiBand can “predict” the self-reported sleep quality and obtain the results reported in Table 2. The balanced accuracy of 5.71% and 17.14% shows that the MiBand can “predict” the self-reported sleep quality worse than a random guesser (20% accuracy for five classes). While this result may improve if fewer classes (e.g., only “good” or “bad” sleep) are considered, or if other devices (e.g., Oura ring or

³For 17 out of the 427 nights of data available in the dataset, users reported awakenings that resulted in multiple sleep segments – and multiple sleep quality labels – during a night. We aggregated such subsegments into a unique segment and used as sleep quality label the weighted average of the labels of the individual segments, weighted proportionally to their duration. Further, we exclude naps reported during the day from the analysis.

⁵<https://www.mi.com/global/miband>

Table 1: List of features extracted from sensor data traces available in the M2Sleep dataset, including their description and method used to extract them.

Feature	Sensor	Description	Method
ACC	ACC	Magnitude of the raw accelerometer signal	$\sqrt{x^2 + y^2 + z^2}$
HR	PPG	Heart rate	nk.ppg_process ⁴
TEMP	TEMP	The raw temperature data	Empatica TEMP raw data
EDA	EDA	The raw EDA signal	Empatica E4 raw data
EDA Peak Epoch		Five peaks or more within a minute	Burch [4]
EDA Storm		Peaks epoch that lasts 10 minutes or more	Sano and Picard [19]
Artifact		Abrupt rise or drop, peak drop or rise quickly, peaks too close	EDArtifact [9]
EDA Filtered		1st order Butterworth low-pass filter, cut-off of 0.6 Hz	EDArtifact [9]
EDA Phasic		Quickly changing component of the raw EDA signal	cvxEDA [10]
EDA Tonic		Slowly changing component of the raw EDA signal	cvxEDA [10]

**Figure 1: Spearman's correlation between non-sensor-based features and self-reported sleep quality. ** p value < 0.001, * p value < 0.01. The results are statistically significant only if p < 0.001, which is the corrected p-value after applying the Bonferroni correction.**

Fitbit [20]) would have been used, it still confirms the existence of a mismatch between measured and perceived sleep quality.

Table 2: Performances of commercial devices for sleep quality (five classes)

User	Accuracy	Balance Accuracy	Recall	Precision
User 1	6.45%	5.71%	6.45%	47.31%
User 2	41.38%	17.14%	41.38%	23.17%

As a next step, we compute the correlation between self-reported sleep quality and each of the non-sensor-based features. We use Pearson's, Spearman's and Kendal's correlation coefficients. Since all measures returned comparable results, we report and discuss here only values obtained using Spearman's measure, shown in Figure 1. We consider a correlation to be statistically significant if the p value is lower than 0.001 (to conform to the Bonferroni correction [2]). The highest correlation values are with sleep duration (0.28) – confirming previous findings [21] –, extroversion (0.23), PSQI (-0.21)⁶ and agreeableness (0.19), all of which are statistically

⁶The lower the PSQI value, the better the sleep quality. A negative correlation thus indicates that the PSQI can help predicting subjective sleep quality.

significant. The high correlation with psychological traits underlines the importance of personality in the subjective assessment of sleep quality.

Lastly, we look at the correlation between self-reported sleep quality and sensor-based features. Figure 2a shows the correlation with statistics obtained using only data from segments *before sleep*. The maximum of the skin temperature (*max TEMP*) was the only feature showing weak correlation (0.26) with self-reported sleep quality. This confirms the results obtained by Kräuchi et al. [15], who reported that a high skin temperature may result in lower sleep latency, which is in turn negatively correlated with subjective sleep quality [6]. Figure 2b shows the correlation coefficients between self-reported sleep quality and statistical features *during sleep*. Acceleration skewness and kurtosis show weak correlation (0.24 and 0.23 respectively). Kurtosis is a measure of the distribution's "tailedness" and high values means narrow bell-shaped curve, and skewness is a measure of the asymmetry of the distribution where higher values means that the distribution is right-skewed. Thus we can derive that having a nearly normal distribution slightly translated towards low values is correlated with higher sleep quality. The 40th percentile and median of tonic EDA also show weak correlation (0.21 and 0.20), indicating that the tonic component of EDA is more suited to capture sleep quality's slowly-changing patterns. Since the correlation with statistical features *after sleep* does not show significant results we do not show that data. For the same reason some features and statistics are omitted in the figures.

5 CONCLUSION

In this paper we highlight that sleep quality measures reported by commercial devices and features computed from sensor data often only weakly correlate with self-reported sleep quality scores. For our analysis we rely on the existing M2Sleep dataset and divide the available data in segments related to time *before*, *during* and *after sleep*. We observe no statistically significant correlation between self-reported sleep quality and physiological responses *after sleep*. Instead, we do observe statistically significant correlation with data collected both *before* and *during* sleep. Specifically, maximum skin temperature before sleep shows the highest correlation among statistical features before sleep. Skewness and kurtosis of the acceleration magnitude, and the 40th percentile of the EDA

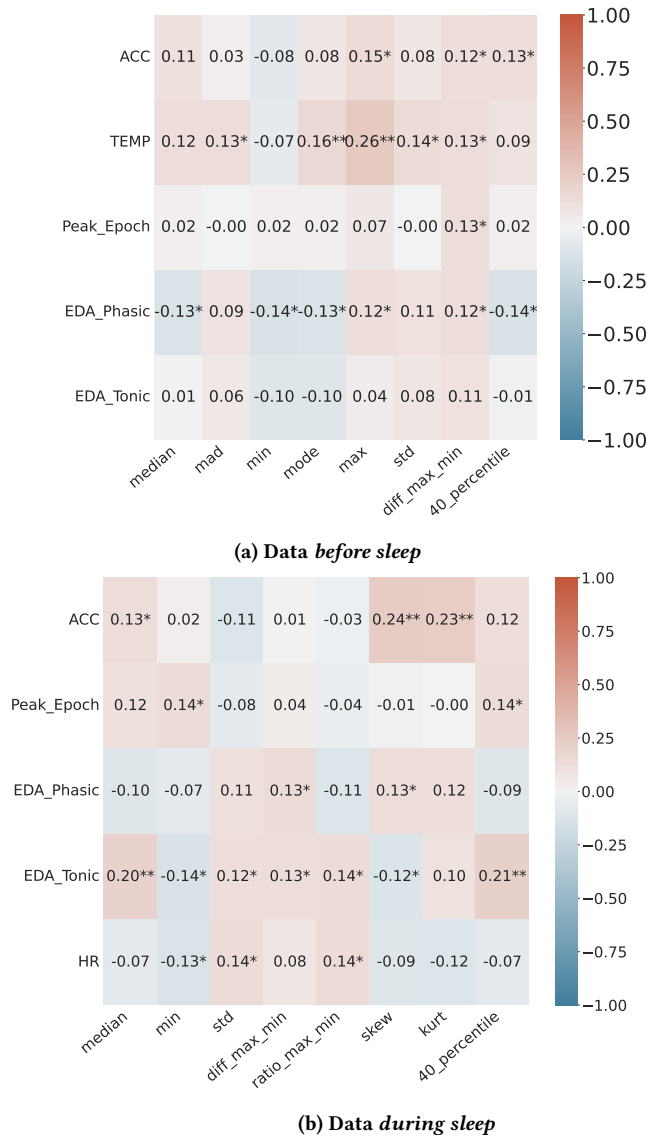


Figure 2: Spearman's correlation coefficient between self-reported sleep quality and sensor-based features extracted (a) before sleep and (b) during sleep. ** p value < 0.001, * p value < 0.01. The results are statistically significant only if $p < 0.001$, which is the corrected p-value after applying the Bonferroni correction.

signal show high correlation during sleep. Our results further highlight that some psychological traits correlate significantly with self-reported sleep quality, suggesting that there may be a bias in the self-reported sleep quality assessments. For this reason future work should focus on using multiple devices to avoid the device bias as well as consider other reliable objective measures (e.g., PSG) to quantify sleep. Taken together, our results call for a cautious consideration of the use of self-reported sleep quality scores as ground-truth labels in sleep quality assessment studies.

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